

CO2-ATL

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Course Data Structures
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Q1)

Definition: -

A Stack is a linear data structure that follows the LIFO (Last in First Out) Principle. The last element inserted into the stack is the first element removed. The two basic operations performed on stack are: -

* Push - Inserts an element into stack

* Pop - Removes the top element from stack.

Initial Stack: -

Array Size = 5

Top = -1

Index: 0 1 2 3 4 5

Stack: [] [] [] [] [] []

top = -1

Execution of Operations: -

Step 1: Push(45)

$$\text{Top} = \text{Top} + 1 = 0$$

Index : 0 1 2 3 4

45				
----	--	--	--	--

$$\text{Top} = 0$$

Step 2: Push(22)

$$\text{Top} = 1$$

Index : 0 1 2 3 4

45	22			
----	----	--	--	--

Step 3: Push(11)

$$\text{Top} = 2$$

Index : 0 1 2 3 4

45	22	11		
----	----	----	--	--

Step 4: Push(77)

$$\text{Top} = 3$$

Index : 0 1 2 3 4

45	22	11	77	
----	----	----	----	--

Step 5: POPC):-

77 removed.

Top = 2

Index : 0 1 2 3 4

45	22	77		
----	----	----	--	--

Step 6: Push(44)

Top = 3

Index : 0 1 2 3 4

45	22	11	44	
----	----	----	----	--

Step 7: Push(33):

Top = 4:

Index : 0 1 2 3 4

45	22	11	44	33
----	----	----	----	----

Step 8: POPC)

33 is removed

Top = 3

Index : 0 1 2 3 4

45	22	11	44	
----	----	----	----	--

Step 9: POPC)

44 is removed

Top = 2

Index : 0 1 2 3 4

45	22	11		
----	----	----	--	--

Step 10: POP()

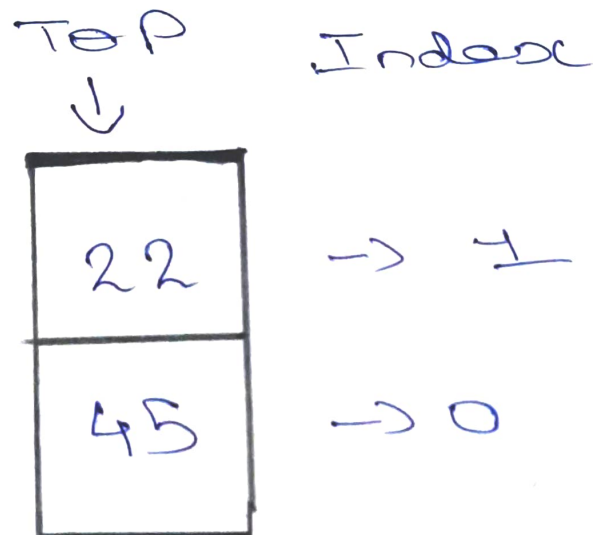
11 is removed:

Top = 1

Index : 0 : 1 2 3 4

45	22			
----	----	--	--	--

Final Stack Diagram:-



Result: -

Peak(Top) Element = 22

Index Position = 1

Time Complexity:-

Push operation = $O(1)$

Pop operation = $O(1)$.

Q2)

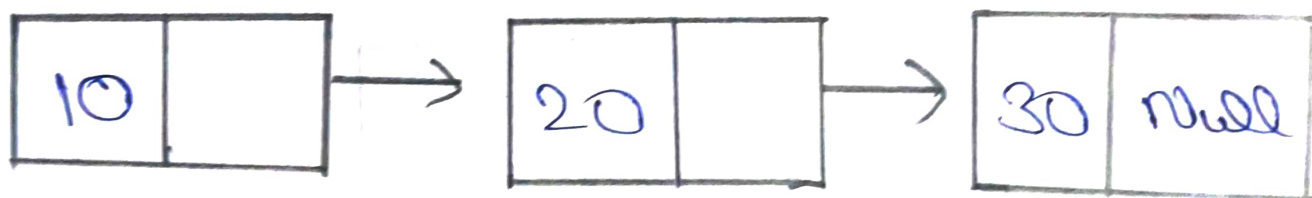
Defn: -

A Queue is a linear data structure that follows the FIFO. The element inserted first is removed first.

The two operations are: -

- * Enqueue - Insert an element at the Rear.
- * Dequeue - Delete an element from the front.

Queue Representation: -



Enqueue operation: -

Defn: -

Enqueue is the process of inserting a new element at Rear of queue.

Before Enqueue (40): -

Front

↓

Rear

10 → 20 → 30 → null

After Enqueue(40)

Front

↓

10 → 20 → 30 → 40 → Null

Explanation: -

- A new node is created
- The data is stored in new node
- The next pointer is initialized to Null
- If queue is empty both front & Rear point to new node.
- Otherwise, the current Rear node is linked to new node.
- The Rear pointer is updated to new node.

Deque Operations: -

Defn: -

Dequeuing is the process of deleting an element from the front of queue.

Before Dequeue: -

Front

↓

10 → 20 → 30 → 40 → Null

After Dequeue: -

Front

↓

20 → 30 → 40 → Null

Explanation: -

1) Check whether the queue is Empty

2) If empty, the queue is in Underflow condition.

3) Otherwise, store the first node in a temporary point

4) Move the front pointer to the next node.

5) Delete the old first node.

6) If queue becomes empty after deletion, set Rear to Null.